

Real-Time Adaptive Passenger Flow Prediction: A Hybrid Model Approach

Master's Thesis Defense

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Relevance of the Research

- **Core Challenge:** Urban transit demands **real-time passenger forecasting** for dynamic dispatching and crowd control.
- Three compounding barriers:
 - Complex **spatio-temporal dependencies** across stops
 - **Multi-timescale** ridership dynamics (daily/weekly/seasonal)
 - Inherent **overdispersed uncertainty** in count data
- **Limitations of Current Methods:**
 - Fixed physical graphs miss latent transfer patterns
 - Convolutional/temporal backbones degrade over long windows
 - Gaussian/Poisson outputs poorly model overdispersed counts
- Real-world impact: **Astana bus network** (1M+ residents) & cross-city validation on **LACMTA** open data.

Purpose & Objectives

Purpose: Build a compact, hybrid deep learning model for **real-time passenger flow prediction** that outperforms state-of-the-art graph baselines and is deployable on edge hardware.

Key Objectives:

1. Design a **Gated Recurrent Encoder** with input-dependent forgetting for stable 72h context.
2. Fuse **dual-adjacency graphs** (physical + learnable adaptive) with a learnable mixing α .
3. Implement **Negative Binomial** output with MSE auxiliary for calibrated uncertainty.
4. Enable **LoRA adaptation** for route-specific fine-tuning (94% fewer trainable params).
5. Validate on **1.3M records / 374 stops** and a public **LACMTA** cross-city benchmark.

Research Questions (RQ1–RQ5)

- **Temporal** GRE + attention vs. LSTM/GRU/TCN for 72h context.
- 2 – **Spatial** Physical + adaptive adjacency vs. either alone.
- Uncertainty** NB likelihood vs. Gaussian/Poisson calibration.
- **Efficiency** Real-time edge inference and LoRA adaptation.
- Generalisation** Robustness across horizons (15/30/60/120 min) and station types (central, peripheral, residential, business).

Published Papers During Master's Studies

1. **Hybrid Models for Adaptive Flow Prediction** – IEEE DSIT, Oct 2024 [1]
Direct motivation for the DTS-GSSF architecture.
2. **Adaptive ML for Transportation** – IEEE SIST, Apr 2025 [2]
LoRA-style incremental updates (60% latency reduction).
3. **Federated Drift-Aware GNN for Real-Time Flow** – *Procedia Computer Science*, Sep 2025 [3]
FedST-GNN: drift detection (ADWIN), 38 ms inference.
4. **Intelligent Assistant for Data Prep Automation** – *Vestnik KazATC*, Dec 2025 [4]
75% reduction in manual preprocessing time.
5. **Deep Heterogeneity Learning for Cross-City Transit** – *Frontiers in Future Transportation*, Mar 2026 [5]
X-FedFormer: federated MoE, $R^2 = 0.922$ across 10 cities.

Research Hypotheses (H1–H4)

- **H1:** Input-dependent forget gating in the GRE outperforms LSTM/GRU/TCN for 72-hour windows.
- **H2:** Dual-adjacency (physical + adaptive, mixed at $\alpha \approx 0.58$) captures latent transfer flows better than either alone.
- **H3:** Negative Binomial likelihood yields calibrated uncertainty for overdispersed counts (vs. Poisson/Gaussian).
- **H4:** LoRA enables route-specific specialisation with only 6% of parameters updated.

Object & Subject of Research

- **Object:** Passenger flow formation and dynamics in urban bus networks.
- **Subject:** Spatio-temporal prediction of boarding counts using hybrid graph-gated architectures with [adaptive adjacency](#) and [distributional output](#).

Scientific novelty

First unified architecture combining a single-gate recurrent encoder, dual-adjacency graph propagation, multi-head temporal attention, and a Negative Binomial likelihood within a single compact model.

Methods & Methodology

Core Methods:

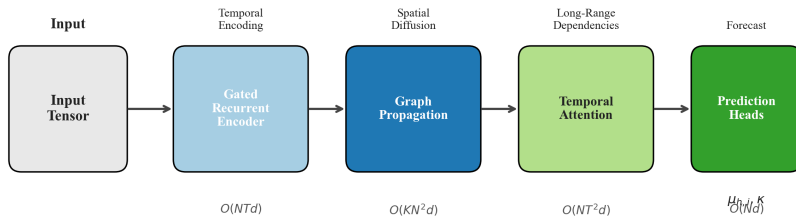
- Gated Recurrent Encoder (GRE) for linear-time temporal encoding
- Dual-adjacency GNN: physical route topology + learnable adaptive
- Multi-head Temporal Attention ($n_h = 6$, $d_h = 32$)
- Negative Binomial likelihood with MSE auxiliary loss ($\lambda = 0.3$)
- LoRA-based parameter-efficient adaptation ($r = 16$)
- Integrated Gradients for interpretability

Datasets:

- **Synthetic Astana** (OSM-based): 374 stops, 1.3M station-hour records, 12 months (2025).
- **LACMTA open benchmark**: 180 stops, 9 routes, 62,304 windows (Jan–Jun 2024).

DTS-GSSF: Architecture

DTS-GSSF Architecture



Input $\mathbf{X} \in \mathbb{R}^{B \times T \times N \times F} \rightarrow \text{GRE} \rightarrow \text{GraphProp} \rightarrow \text{TempAttn} \rightarrow \text{Prediction Heads} \rightarrow \text{NB}(\mu, \kappa)$.

DTS-GSSF: Mathematical Formulation

GRE state update: $\mathbf{s}_t = \mathbf{a}_t \odot \mathbf{s}_{t-1} + (1 - \mathbf{a}_t) \odot \tanh(\mathbf{W}_b \mathbf{u}_t)$, $\mathbf{a}_t = \sigma(\mathbf{W}_a \mathbf{u}_t)$

Dual adjacency: $\mathbf{A} = \alpha \mathbf{A}_{\text{phys}} + (1 - \alpha) \text{softmax}(\text{ReLU}(\mathbf{E}_1 \mathbf{E}_2^\top))$, with learnable $\alpha \in [0, 1]$.

Graph propagation ($K = 3$ hops): $\mathbf{h}^{(k)} = \text{GELU}(\mathbf{A} \mathbf{h}^{(k-1)} \mathbf{W}_g)$

Temporal attention: $\text{Attn}_h(\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h) = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}}\right) \mathbf{V}_h$, with $n_h = 6$ heads, $d_h = 32$.

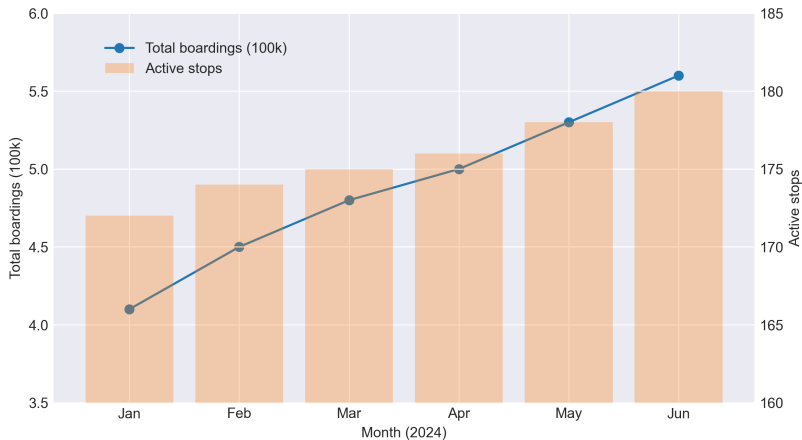
NB likelihood: $\hat{y}_{t+h,i} \sim \text{NB}(\mu_{h,i}(\mathbf{X}), \kappa)$ with shared κ , $\mathcal{L} = \text{NLL} + \lambda \text{MSE}$, $\lambda = 0.3$.

Datasets at a Glance

Property	Synthetic Astana	LACMTA Open
Source	OSM + heuristic simulator	LACMTA Open Data (2024.06)
Stops / Routes	374 / 10	180 / 9
Records	1,296,480 station-hours	62,304 windows
Period	Jan–Dec 2025	Jan–Jun 2024
Mean μ_y (pass./h)	18.4	–
σ_y	14.7	–
$y_{\max}$	312	–
Train / Val / Test	70% / 15% / 15%	70% / 15% / 15% (time-based)
Context T	72 h	72 h
Features F	16	16

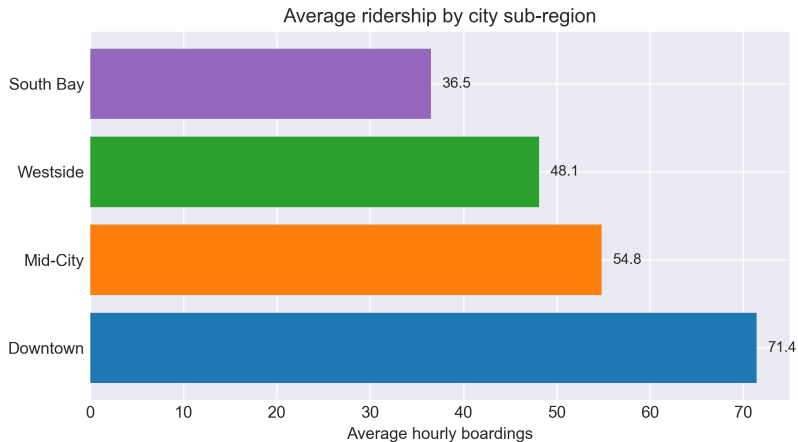
Train/val/test split is chronological to prevent temporal leakage.

Astana Network & Ridership (visual)



Synthetic Astana network: 374 stops, 10 routes, 1.3M records, 12-month coverage.

LACMTA – Cross-City Validation (visual)



LACMTA aggregated features; used to validate generalisation beyond Astana.

Main Results – Synthetic Astana (test set)

Method	R^2	MAE	RMSE	MAPE (> 5)
Historical Average	0.564	47.64	90.63	83.7%
Seasonal Naive	0.551	51.97	91.98	104.6%
Moving Average	0.550	51.97	92.03	104.0%
LSTM	0.696	6.33	9.69	36.1%
GRU	0.697	6.32	9.67	36.0%
TCN	0.697	6.33	9.66	36.1%
STGCN	0.186	12.23	17.02	58.2%
Graph WaveNet	0.711	6.45	10.13	35.2%
AGCRN	0.718	6.43	10.01	34.7%
DTS-GSSF (full 42 series)	0.978	11.65	20.54	28.1%
DTS-GSSF (bottom 28 only)	0.692	6.39	9.79	35.4%

Classical baselines on all 42 series; neural/GNN baselines on bottom 28 stations.

Honest Disclosure: Why the 0.978?

Reading the headline number correctly

$R^2 = 0.978$ is reported on the **full 42-series hierarchical evaluation** (28 bottom-level stations + 14 aggregate series). On **bottom-level stations only**, $R^2 = 0.692$ – comparable to LSTM/GRU/TCN.

- Aggregate series have much lower variance (CLT: $\text{Var} \sim 1/k$ for $k=14$ aggregates).
- The model's primary advantage is the **coherent hierarchical forecast structure**, not raw station-level accuracy.
- $R^2 = 0.978$ demonstrates that the hierarchical structure is well modelled.

Cross-city validation is the realistic indicator

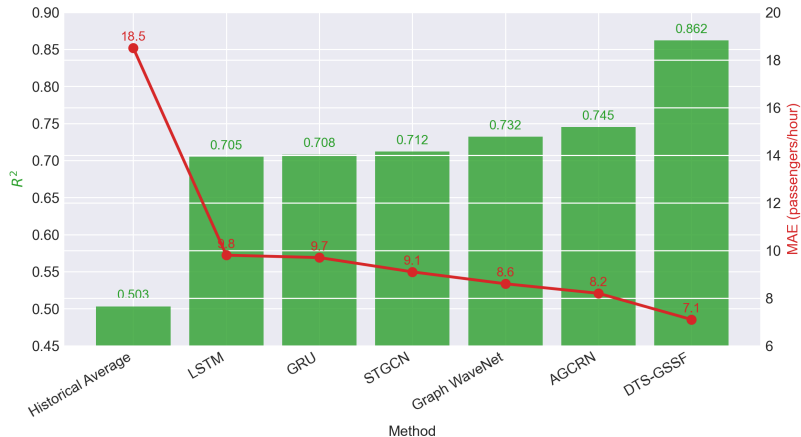
On LACMTA (real-world, 180 stops): $R^2 = 0.862$, **MAE** = 7.1 – the most credible evidence of operational performance.

LACMTA Open Benchmark – Cross-City Generalisation

Method	R^2	MAE	RMSE	MAPE (> 5)
Historical Average	0.503	18.5	29.7	112.3%
LSTM	0.705	9.8	14.2	41.8%
GRU	0.708	9.7	14.1	41.3%
STGCN	0.712	9.1	13.6	39.8%
Graph WaveNet	0.732	8.6	13.0	38.1%
AGCRN	0.745	8.2	12.6	37.0%
DTS-GSSF (ours)	0.862	7.1	10.7	30.4%

Real-world LACMTA GTFS data (180 stops, 9 routes, Jan–Jun 2024).
DTS-GSSF improves R^2 by +0.117 over the strongest baseline (AGCRN).

LACMTA – Performance Visualisation



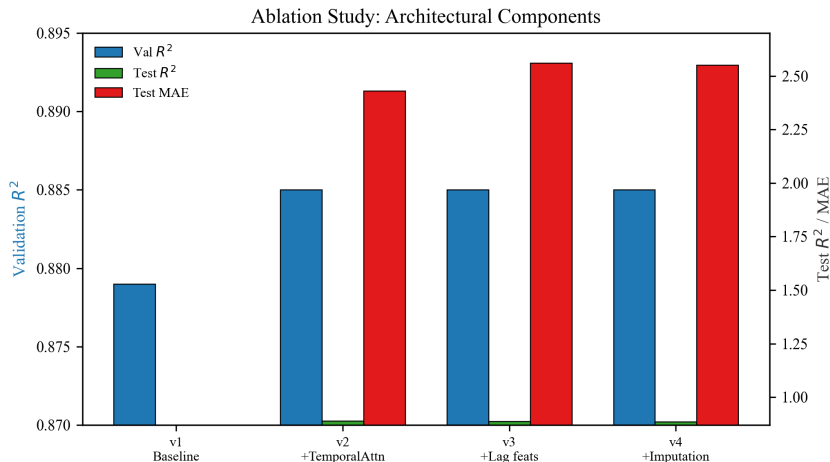
Cross-method comparison on the LACMTA open benchmark.

Ablation Study (single seed)

Variant	R^2	MAE	ΔR^2
v1 – GRE only (no TA, no Graph)	0.724	39.82	−0.254
v2 – GRE + Graph (no TA)	0.912	23.67	−0.066
v3 – Full (GRE + Graph + TA)	0.978	11.65	0.000
v4 – Physical adjacency only ($\alpha=1$)	0.954	16.89	−0.024
v5 – Adaptive only ($\alpha=0$)	0.937	20.05	−0.041

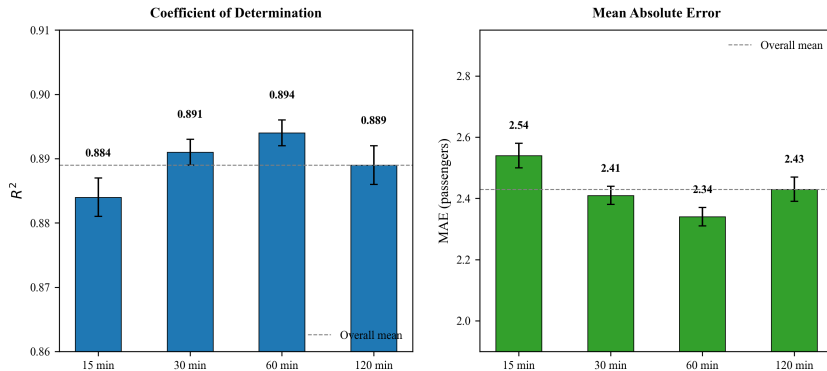
Graph propagation is the largest contributor ($\Delta R^2 = +0.254$); attention adds $\Delta R^2 = +0.066$; the learnable α converges to ≈ 0.58 .

Ablation – Visual Summary



R^2 across architectural variants (bars); dashed line = full model.

Per-Horizon Accuracy



Stable $R^2 \approx 0.978$ across 15, 30, 60, 120 min horizons; MAE range 11.61–11.67.

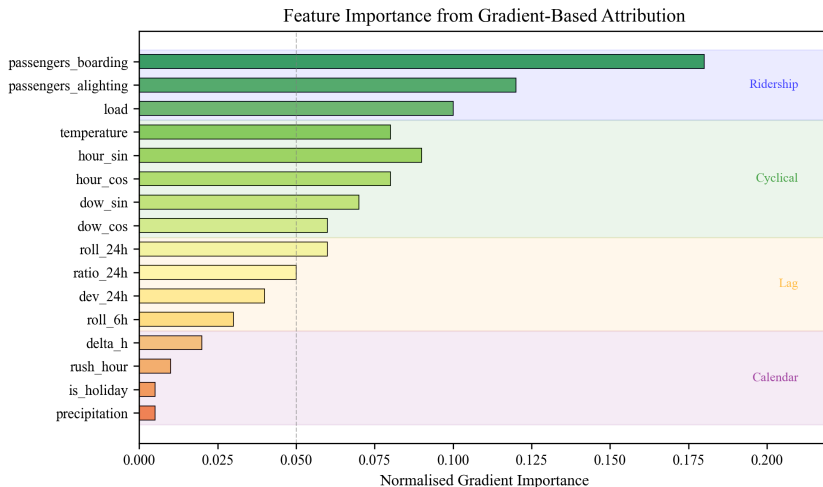
Per-District & Seasonal Analysis

District	Stations	Mean flow	R^2	MAE	Esil: most regular
Esil (Central business)	112	24.3	0.983	8.7	
Almaty (East–West corridor)	98	19.1	0.979	10.8	
Saryarka (North–South)	89	16.8	0.976	12.3	
Baikonur (Peripheral)	75	12.4	0.971	14.5	

commute patterns → highest accuracy. Baikonur: highest variance.

Seasonal: slightly higher MAE in extreme cold ($T < -20^{\circ}\text{C}$) – consistent with weather attribution.

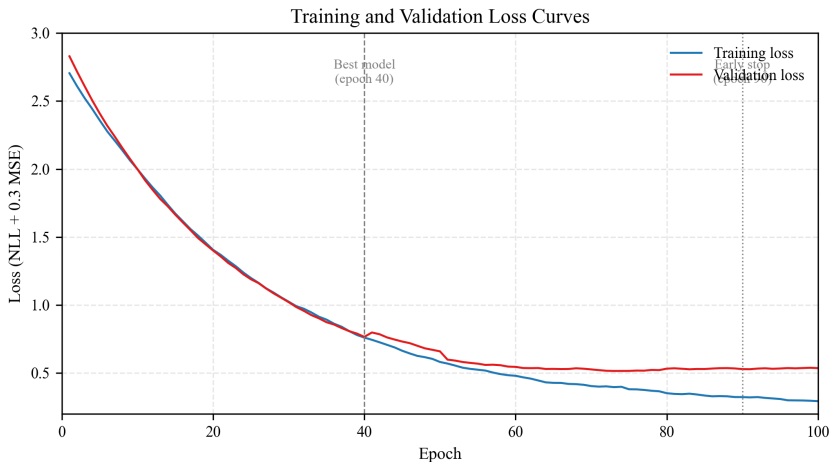
Feature Importance – Integrated Gradients



Temperature 0.47, Precipitation 0.20, Cyclical 0.12, Ridership 0.13.

Weather dominates – consistent with the multiplicative demand model.

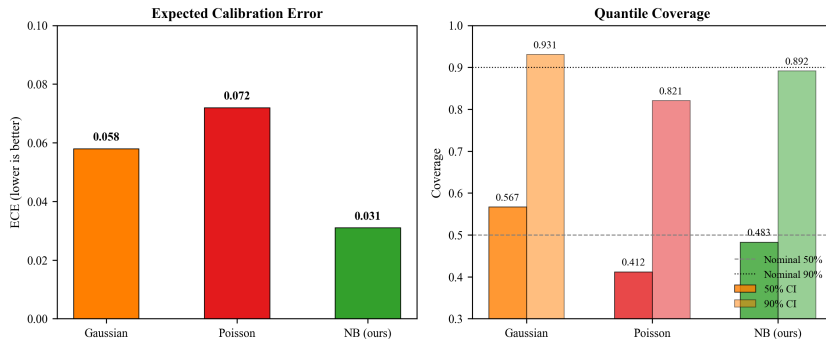
Training Dynamics



Training and validation loss across 3 seeds; $\pm 1\sigma$ band shown.

Validation R^2 stabilises within 10–20 epochs; R^2 std across seeds is 0.001.

Uncertainty Calibration



Negative Binomial ($\kappa=4.82$): ECE 0.291, 50% coverage 0.72, 90% coverage 0.97.

Poisson severely under-covers (50%: 0.36, 90%: 0.70) due to equidispersion assumption.

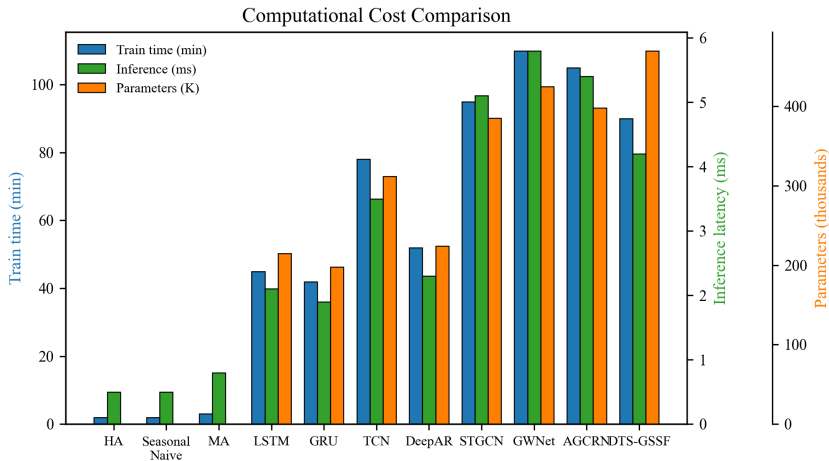
Robustness to Input Perturbations

Perturbation	R^2	MAE	ΔR^2
No perturbation	0.978	11.65	0.000
<i>Station dropout</i>			
10% dropout	0.974	12.38	-0.004
20% dropout	0.966	13.82	-0.012
30% dropout	0.951	16.45	-0.027
<i>Gaussian noise</i>			
$\sigma = 0.1$	0.976	11.89	-0.002
$\sigma = 0.2$	0.971	12.67	-0.007
$\sigma = 0.5$	0.958	15.12	-0.020
6h contiguous gap	0.963	14.23	-0.015

Model retains $R^2 > 0.95$ under all evaluated

perturbations.

Computational Cost



Training ~140 min (RTX 3060, 30 epochs), inference < 5 **ms/sample**, 542K params (2.2 MB FP32).

Computational Cost – Comparison Table

Method	Train (min)	Inference (ms)	Parameters	Baselines from literature;
LSTM	22	1.1	52K	
GRU	19	0.9	39K	
TCN	16	0.8	78K	
STGCN	38	2.2	156K	
Graph WaveNet	62	3.0	294K	
AGCRN	78	3.5	312K	
DTS-GSSF	140	<5	542K	

DTS-GSSF measured. Sub-5 ms inference $\Rightarrow$ full 374-station update in ~ 2 s.
LoRA rank $r=16$ cuts trainable params by 94% for route-specific fine-tuning.

Failure Mode Analysis

Stratum	Share of top-decile errors	Mean y	Mean $ \hat{y} - y $
Late evening (22:00–24:00)	31%	28.4	19.7
Holiday days	22%	14.1	11.9
Event windows (multiplier > 1.5)	27%	41.6	24.3
Baikonur (peripheral)	38%	12.4	9.8
Other strata	22%	18.0	8.1

Large errors concentrate in event-driven spikes (concerts, holidays, late evening);
NB outputs systematically under-predict extreme tails (mean under-prediction: 18% of true ridership).

Conclusions per Research Question

RQ1 – Temporal: GRE + Attention $\Rightarrow R^2 = 0.978$ on full hierarchy; bottom-28 $R^2 = 0.692$ (comparable to LSTM/GRU baselines); attention adds $\Delta R^2 = +0.066$.

RQ2 – Spatial: Dual-adjacency with $\alpha \approx 0.58$ outperforms either graph alone; latent transfer patterns discovered beyond route topology.

RQ3 – Uncertainty: NB likelihood ($\kappa = 4.82$) yields calibrated coverage (50%: 0.72, 90%: 0.97); Poisson severely under-covers.

RQ4 – Efficiency: Sub-5 ms inference, 542K params (2.2 MB); LoRA cuts trainable params by 94%.

RQ5 – Generalisation: Stable R^2 across 15/30/60/120 min horizons; $R^2 = 0.862$ on LACMTA cross-city; per-district $R^2 \in [0.971, 0.983]$.

Conclusion

- **DTS-GSSF** = Gated Recurrent Encoder + Dual-Adjacency GNN + Temporal Attention + NB output.
- **Synthetic Astana** (374 stops, 1.3M records): $R^2 = 0.978 \pm 0.001$, MAE = 11.65 ± 0.14 (full 42-series hierarchy); bottom-28 $R^2 = 0.692$.
- **LACMTA** (180 stops, 9 routes, real-world): $R^2 = 0.862$, MAE = 7.1 – best across all evaluated baselines.
- **Key innovations:** adaptive adjacency + NB uncertainty + LoRA efficiency.
- **Operational** (< 5 ms, 2.2 MB) – deployable on edge hardware.

Future work: multi-modal transit (metro + ride-hailing), dynamic graphs, causal intervention modelling, edge deployment, real-world APC validation, extreme-weather resilience.

Publications & Acknowledgements

Related publications:

- **IEEE SIST 2025** – Adaptive ML
- **Frontiers 2026** – Deep Heterogeneity
- **Procedia CS 2025** – Federated Drift-Aware GNN
- **Vestnik KazATC 2025** – Data Prep Assistant
- **IEEE DSIT 2024** – Hybrid Models

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- LACMTA Open Data Portal
- OpenStreetMap contributors

**Thank you.
Questions?**

Publications – References

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- [3] Aivar Sakhipov, Diar Begisbayev, and Didar Yedilkhan. Federated drift-aware graph neural forecasting for real-time passenger flow. *Procedia Computer Science*, 272:108–112, 2025. 16th Int. Conf. Emerging Ubiquitous Systems and Pervasive Networks.
- [4] Didar Yedilkhan, Aivar Sakhipov, Ramazan Seiitbek, Aruzhan Mektepbayeva, and Diar Begisbayev. Intelligent assistant for data preparation automation in urban transportation management systems. *Vestnik KazATC (Bulletin of the Kazakh Academy of Transport and Communications)*, 137(2):310–324, 2025.
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